



Stockholm
University

Department of Meteorology

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Ocean Gyre Circulation Sverdrup and Munk Solutions

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June 1, 2026

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Method

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Result, The Sverdrup solution

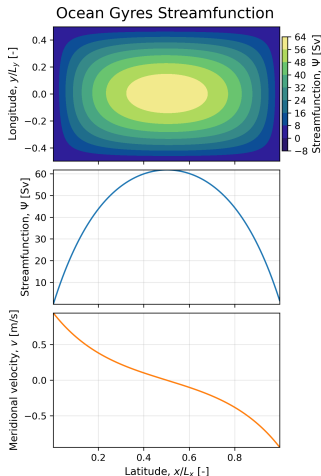


Figure: Streamfunction for the no-drag f -plane configuration. The circulation remains symmetric with no β -effect and without western intensification.

Result, The Sverdrup solution

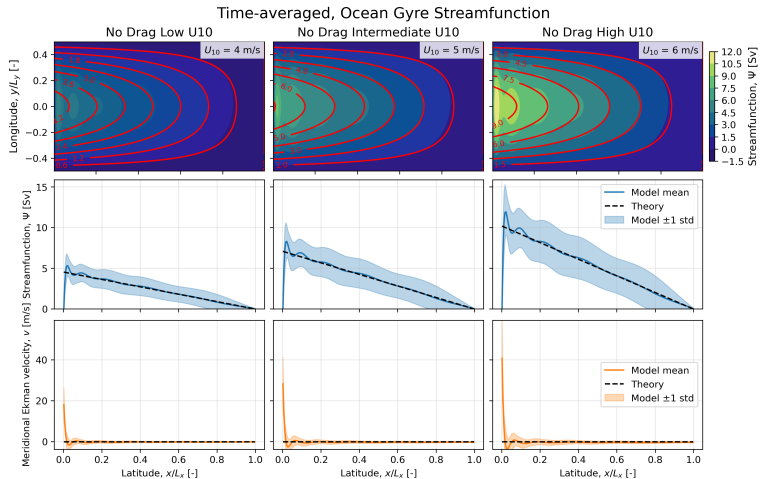


Figure: Streamfunction for the no-drag β -plane configuration.

Result, The Sverdrup solution

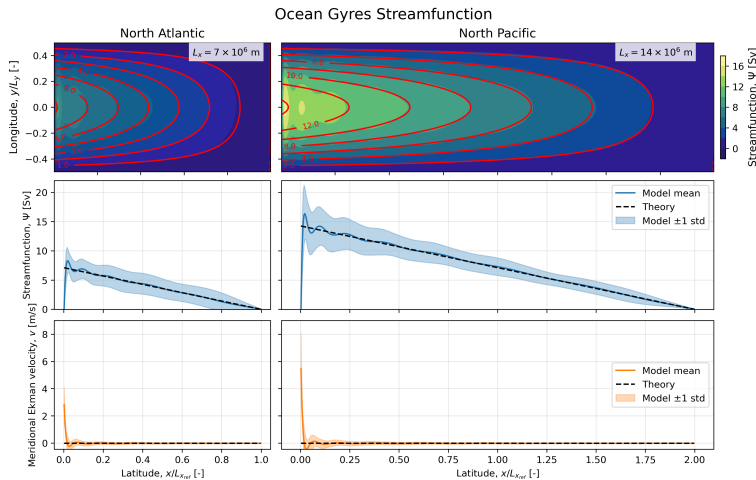


Figure: Streamfunction for the no-drag β -plane configuration with varying domain size.

Result, The Munk solution

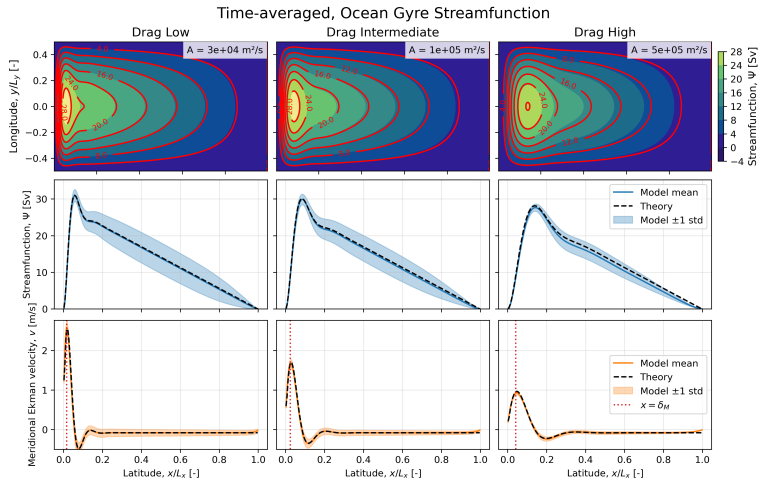


Figure: Streamfunction for the Laplace-viscous β -plane configuration.

Result, The Munk solution

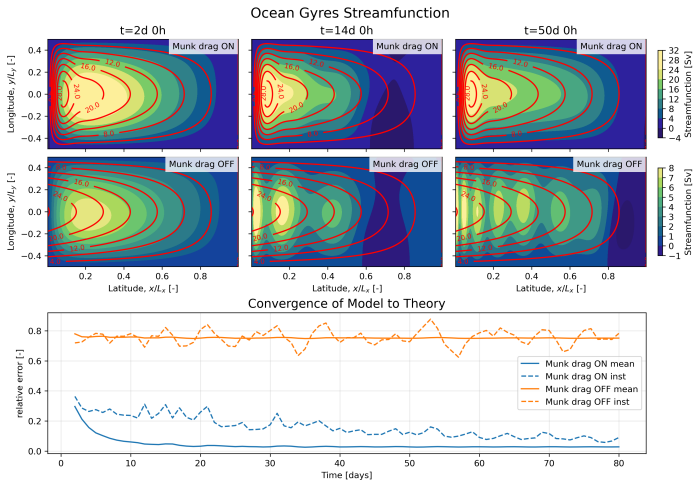


Figure: Streamfunction for the Laplace-viscous β -plane configuration.

Discussion, Sverdrup versus Munk

Conclusion